



# Technical Service Bulletin

Technical Service Bulletin: TSB200067

Released Date: 24-Mar-2021

Fault Code 285/286 Multiplexing

## Fault Code 285/286 Multiplexing

### Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

### Contents

#### Product Affected

- Industrial
  - QSK19 CM850 MCRS
  - QSK19 CM2150 MCRS
  - QST30 CM2350 T101
  - QSK38 CM850 MCRS
  - QSK38 CM2150 K106
  - QSK38 CM2350 K125
  - QSK50 CM850 MCRS
  - QSK50 CM2150 K107
  - QSK50 CM2350 K108
  - QSK60 CM850 MCRS
  - QSK60 CM2150 MCRS
  - QSK60 CM2350 K116
  - QSK60 CM2350 K117
  - QSK60 CM2350 K121
  - QSK60 CM2350 K135
  - QSK60 CM2350 K136
  - QSK78 CM2350 K126
  - QSK78 CM2350 K137
  - QSK95 CM2350 K111
  - QSK95 CM2350 K113
  - QSK95 CM2350 K128M
- Power Generation
  - QSK19 CM850 MCRS
  - QSK19 CM2150 MCRS
  - QSK38 CM850 MCRS
  - QSK38 CM2150 K106

- QSK50 CM850 MCRS
- QSK50 CM2150 K107
- QSK60 CM850 MCRS
- QSK60 CM2150 MCRS
- QSK95 CM2350 K111

### **Description of Change**

This document announces a calibration correction to enable the display and description of Fault Code (FC) 285 and 286 on the Fault Data screen on Insite.

### **Reason for Change**

In the past, the root cause of Fault Codes 285 or 286 has been found to not be viewable on Insite due to a software feature not being enabled on High Horse Power (HHP) calibrations. Fault Codes FC285/FC286 are generated due to a missing J1939 message from the Original Equipment Manufacturer (OEM).

Calibrations have a feature that will display which missing J1939 multiplex message is causing the 285/286 fault code in the Advanced ECM Data tab of Insite; this feature was turned off on various calibrations.

### **Service Instructions**

See Figures 1 and 2 below from Insite.

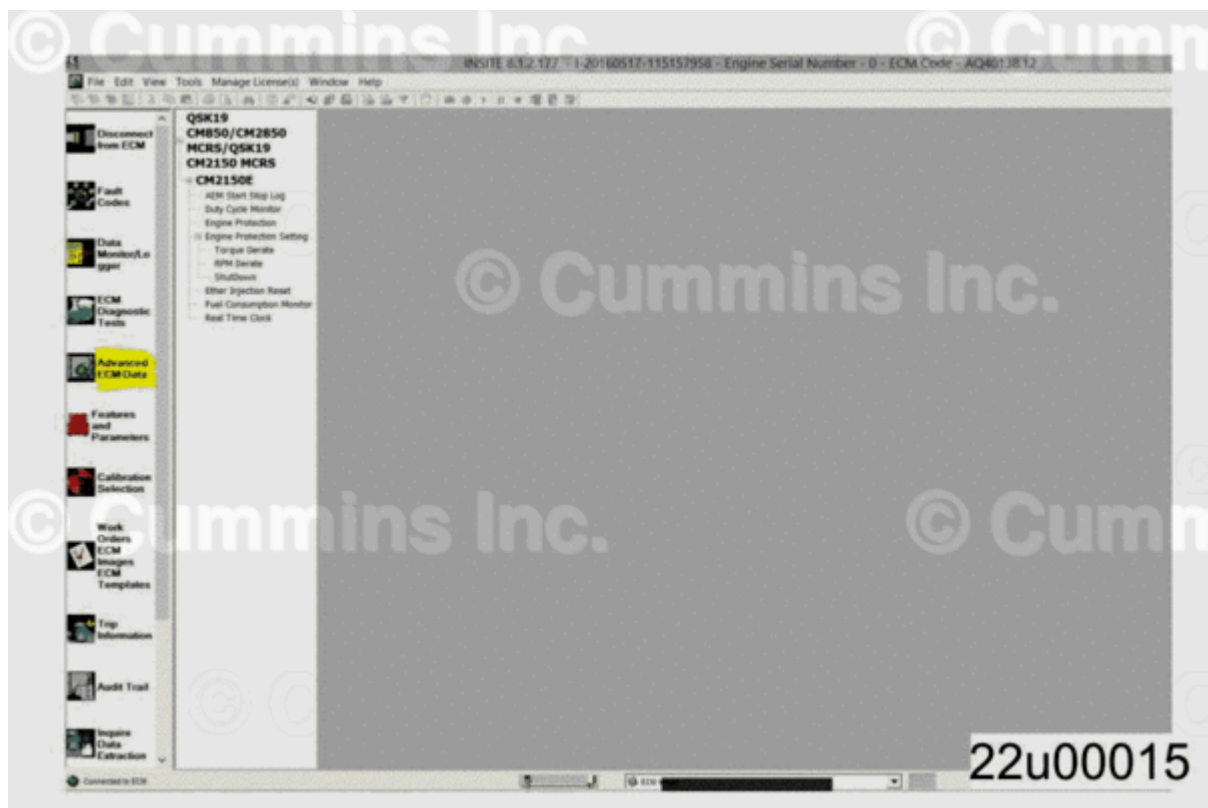


Figure 1, Before Calibration Correction - Insite Is Not Displaying the SAE J1939 Fault Data Screen

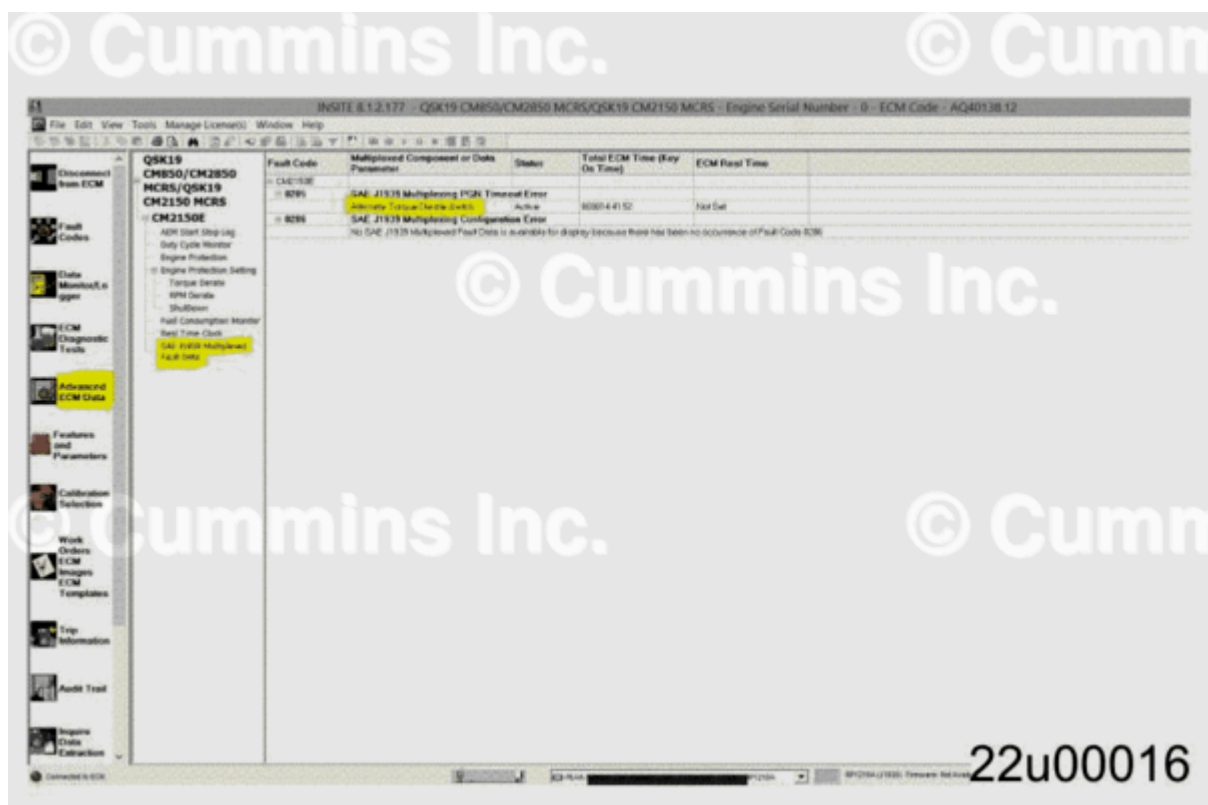


Figure 2, After Calibration Correction - Showing Insite Advanced ECM Data Screen with SAE J1939 Fault Driving the 285 Fault

**ECM Compatibility**

Make sure that the ECM calibrations installed on the engine are listed in the tables below. Please note that all ECM calibrations released prior to this software release will not support these fault codes.

**Industrial**











|            |         |  |  |
|------------|---------|--|--|
| AQ40312.01 | FR04511 |  |  |
| AQ40313.01 | FR04549 |  |  |
| AQ40314.01 | FR04556 |  |  |
| AQ40315.01 | FR04509 |  |  |
| AQ40316.01 | FR04512 |  |  |
| AQ40317.02 | FR04514 |  |  |
| AQ40318.01 | FR04551 |  |  |
| AQ40319.01 | FR04514 |  |  |
| AQ40320.01 | FR04564 |  |  |
| AQ40322.02 | FR04549 |  |  |
| AQ40326.00 | FR04564 |  |  |
| AQ40330.00 | FR04565 |  |  |
| AQ40334.00 | FR04549 |  |  |

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|            |         |  |  |
|------------|---------|--|--|
| AQ60707.03 | FR60210 |  |  |
| AQ60713.10 | FR06780 |  |  |
| AQ60714.10 | FR06855 |  |  |
| AQ60718.03 | FR60237 |  |  |
| AQ60719.03 | FR60238 |  |  |
| AQ60720.03 | FR60239 |  |  |
| AQ60721.04 | FR60242 |  |  |
| AQ60722.09 | FR60240 |  |  |
| AQ60723.08 | FR06735 |  |  |
| AQ60736.08 | FR06869 |  |  |
| AQ60737.08 | FR06904 |  |  |
| AQ60782.03 | FR06843 |  |  |
| AQ60783.03 | FR06844 |  |  |
| AQ60784.03 | FR06842 |  |  |
| AQ60802.03 | FR60240 |  |  |
| AQ60878.01 | FR06779 |  |  |
| AQ60905.00 | FR06853 |  |  |
| AQ60906.00 | FR06853 |  |  |
| AQ60907.00 | FR06853 |  |  |
| AQ60908.00 | FR06854 |  |  |
| AQ60909.00 | FR06855 |  |  |
| AQ60910.00 | FR06855 |  |  |
| AQ60956.01 | FR60643 |  |  |
| AQ60975.00 | FR60649 |  |  |
| AQ60976.00 | FR60650 |  |  |
| AQ60977.00 | FR60651 |  |  |
| AQ60978.00 | FR60653 |  |  |









|            |         |  |  |
|------------|---------|--|--|
| AQ60939.01 | FR06859 |  |  |
| AQ60940.01 | FR06859 |  |  |
| AQ60941.02 | FR06794 |  |  |
| AQ60942.02 | FR06901 |  |  |









|            |         |  |  |
|------------|---------|--|--|
| AQ60916.01 | FR06895 |  |  |
| AQ60917.01 | FR06895 |  |  |
| AQ60918.02 | FR06897 |  |  |
| AQ60919.01 | FR06897 |  |  |
| AQ60920.01 | FR06907 |  |  |
| AQ60921.01 | FR06898 |  |  |
| AQ60922.01 | FR06898 |  |  |
| AQ60923.01 | FR06898 |  |  |
| AQ60924.01 | FR06898 |  |  |
| AQ60925.01 | FR06908 |  |  |
| AQ60926.01 | FR06908 |  |  |
| AQ60927.01 | FR06899 |  |  |
| AQ60928.01 | FR06900 |  |  |
| AQ60929.01 | FR06938 |  |  |
| AQ60930.01 | FR06938 |  |  |
| AQ60931.01 | FR60323 |  |  |
| AQ60932.01 | FR60323 |  |  |
| AQ60933.01 | FR60408 |  |  |
| AQ60946.01 | FR06796 |  |  |
| AQ60947.01 | FR06896 |  |  |
| AQ60948.00 | FR06896 |  |  |
| AQ60949.00 | FR06830 |  |  |

| QSK78 CM2350 |           | QSK95 CM2350 |           |
|--------------|-----------|--------------|-----------|
| ECM Code     | FR Option | ECM Code     | FR Option |
| HH60002.09   | FR60315   | EY80001.15   | FR07327   |
| HH60003.07   | FR60315   | EY80002.06   | FR07330   |
| HH60005.08   | FR60451   | EY80004.15   | FR07440   |
| HH60006.05   | FR60451   | EY80007.10   | FR07327   |
| HH60007.05   | FR60315   | EY80010.09   | FR07501   |
|              |           | EY80011.10   | FR07502   |
|              |           | EY80012.07   | FR07501   |
|              |           | EY80013.07   | FR07501   |
|              |           | EY80014.07   | FR07502   |
|              |           | EY80015.03   | FR07498   |
|              |           | EY80016.00   | FR07538   |

## Power Generation

| QSK19 CM850/2150 |           |  | QSK38 CM850/2150 |           |
|------------------|-----------|--|------------------|-----------|
| ECM Code         | FR Option |  | ECM Code         | FR Option |
| AQ40117.08       | FR04528   |  | AQ60306.08       | FR06757   |
| AQ40125.06       | FR04529   |  | AQ60307.08       | FR06758   |
| AQ40268.06       | FR04609   |  | AQ60536.05       | FR60095   |
| AQ40302.03       | FR04715   |  | AQ60708.03       | FR60211   |
| AQ40303.03       | FR04716   |  | AQ60709.03       | FR60212   |
| AQ40323.02       | FR04770   |  | AQ60969.00       | FR60652   |

| QSK50 CM850/2150 |           |  | QSK60 CM850/2150 |           |
|------------------|-----------|--|------------------|-----------|
| ECM Code         | FR Option |  | ECM Code         | FR Option |
| AQ60290.07       | FR06762   |  | AQ60256.07       | FR06771   |
| AQ60291.07       | FR06763   |  | AQ60257.07       | FR06772   |
| AQ60533.06       | FR60096   |  | AQ60468.05       | FR06987   |
| AQ60703.06       | FR60165   |  | AQ60760.02       | FR60313   |
| AQ60711.05       | FR60166   |  | AQ60814.01       | FR60355   |
| AQ60746.02       | FR60306   |  | AQ60816.01       | FR60354   |
| AQ60895.02       | FR60535   |  | AQ60820.01       | FR60486   |
| AQ60896.02       | FR60536   |  | AQ60821.02       | FR60487   |
| AQ60897.01       | FR60537   |  | AQ60829.02       | FR60490   |
| AQ60954.00       | FR60545   |  | AQ60830.02       | FR60491   |

| QSK95 CM2350 K111 |           |
|-------------------|-----------|
| ECM Code          | FR Option |
| EW80001.16        | FR07370   |
| EW80002.17        | FR07372   |
| EW80003.12        | FR07368   |
| EW80004.14        | FR07372   |
| EW80005.13        | FR07369   |
| EW80006.13        | FR07371   |
| EW80007.13        | FR07371   |
| EW80008.13        | FR07369   |
| EW80010.12        | FR07368   |

| QSK95 CM2350 K111 |           |
|-------------------|-----------|
| ECM Code          | FR Option |
| EW80011.12        | FR07370   |
| EW80017.07        | FR07511   |
| EW80018.07        | FR07511   |
| EW80019.06        | FR07525   |
| EW80020.06        | FR07526   |
| EW80022.02        | FR07369   |
| EW80023.01        | FR07527   |
| EW80024.01        | FR07528   |
| EW80025.01        | FR07529   |
| EW80026.01        | FR07530   |
| EW80027.01        | FR07531   |
| EW80028.01        | FR07532   |
| EW80029.01        | FR07533   |
| EW80030.01        | FR07534   |
| EW80031.01        | FR07529   |
| EW80033.01        | FR07548   |
| EW80034.01        | FR07549   |
| EW80035.01        | FR07550   |
| EW80036.01        | FR07551   |
| EW80037.01        | FR07552   |
| EW80038.01        | FR07553   |
| EW80039.01        | FR07554   |
| EW80040.01        | FR07555   |
| EW80041.01        | FR07556   |
| EW80042.01        | FR07557   |
| EW80043.01        | FR07558   |
| EW80044.01        | FR07585   |
| EW80045.01        | FR07372   |

## Document History

| Date      | Details                  |
|-----------|--------------------------|
| 2020-3-19 | Module Created           |
| 2021-3-23 | Added QSK95 CM2350 K111. |

